An algebraic approach to the Riemann-Roch theorem and the arithmetic theory of function fields

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Abstract.

The Riemann-Roch theorem is usually cast in the language of analysis and geometry, but it can also be regarded as a statement about the dimensions of algebraic extensions of arbitrary fields. The aim of this paper is to prove the Riemann-Roch theorem in this purely algebraic fashion and then discuss the number-theoretic significance of this approach. The first section of the paper introduces the language of valuations and places that is to be used in the proof and culminates with the proof itself; the second part discusses the significance of this point of view and how it leads to an arithmetic theory for function fields not unlike the one for number fields. Our exposition in the first part is inspired by the treatment in Stichtenoth's [ST] and Iwasawa's [IW] books, and the second part follows Rosen [RO].

1 The Riemann-Roch Theorem.

1.1 Valuations, Valuation Rings and Places.

We shall develop the algebraic theory leading to the proof of the Riemann-Roch theorem in the most general setting possible. In this respect, we emulate the approach initiated by Artin, Hasse, Schmidt and Weil in the first half of the twentieth century which superseded the studies of Dedekind, Kronecker and Weber in the nineteenth century (they restricted their attention to the complex numbers). Therefore, in all that follows $\mathbb{K}$ will denote an arbitrary field, unless otherwise noted.

Definition 1.1. *A function field $\mathbb{F}/\mathbb{K}$ of one variable over $\mathbb{K}$ is an algebraic extension $\mathbb{F} \supseteq \mathbb{K}$ such that $\mathbb{F}$ is a finite extension of $\mathbb{K}(x)$, where x is an indeterminate (it's of course equivalent to regard it as an element of $\mathbb{F}$ that is transcendental over $\mathbb{K}$).*

The algebraic closure of $\mathbb{K}$ in $\mathbb{F}$ will be denoted by $\tilde{\mathbb{K}}$ and will be called the *field of constants* of $\mathbb{F}/\mathbb{K}$; note that $\tilde{\mathbb{K}}/\mathbb{K}$ is also a function field over $\mathbb{K}$. If $\mathbb{K}$ and $\tilde{\mathbb{K}}$ coincide, then $\mathbb{K}$ is called *algebraically closed with respect to* $\mathbb{F}$; equivalently, we may say that $\mathbb{K}$ is the *full constant field of* $\mathbb{F}$.

We will now introduce some standard language from Commutative Algebra. We begin with

Definition 1.2. *A valuation ring of the function field $\mathbb{F}/\mathbb{K}$ is a ring $\mathbb{K} \subsetneq \mathcal{O} \subsetneq \mathbb{F}$ such that for all $z \in \mathbb{F}$ with $z \notin \mathcal{O}$, we have $z^{-1} \in \mathcal{O}$.*

We shall need a few properties of valuation rings, which are covered in most books on Commutative Algebra, e.g. [AM], chapters 5 and 9; we shall include their proofs here for completeness.

Theorem 1.1. *Let $\mathcal{O}$ be a valuation ring of the function field $\mathbb{F}/\mathbb{K}$. Then*

(a) $\mathcal{O}$ is a local ring, its unique maximal ideal $\mathfrak{m}$ being the complement of the group of units $\mathcal{O}^\times$ of $\mathcal{O}$.

(b) If $\tilde{\mathbb{K}}$ is the field of constants of $\mathbb{F}/\mathbb{K}$, then $\tilde{\mathbb{K}} \subseteq \mathcal{O}$ and $\tilde{\mathbb{K}} \cap \mathcal{O} = \{0\}$.

Proof.

(a) We claim that the set $\mathfrak{m} = \mathcal{O} - \mathcal{O}^\times$ of non-units is an ideal of $\mathcal{O}$; if this is true, then it follows immediately that $\mathfrak{m}$ is maximal. Indeed, if $x \in \mathfrak{m}, y \in \mathcal{O}$, then $xy \in \mathfrak{m}$, or otherwise x would be a unit. For additive closure, note that given two non-zero elements x, y of $\mathfrak{m}$, $x/y \in \mathcal{O}$ or $y/x \in \mathcal{O}$; assume without loss of generality the former. Then, $1 + x/y \in \mathcal{O}$ and thus $x + y = y(1 + x/y) \in \mathcal{O}$ by the above. This completes the proof that $\mathfrak{m}$ is a (maximal) ideal of $\mathcal{O}$.

(b) Let $z \in \tilde{\mathbb{K}}$ and assume that $z \notin \mathcal{O}$. Then, we must have $z^{-1} \in \mathcal{O}$ and since z^{-1} is algebraic over K, there is a polynomial relation

$$a_m z^{-m} + \cdots + a_1 z^{-1} + 1 = 0,$$

whence

$$z = -(a_m z^{-m+1} + \cdots + a_1) \in \mathcal{O},$$

hence $z \in \mathcal{O}$, which is absurd. Therefore, $z \in \mathcal{O}$. The statement $\tilde{\mathbb{K}} \cap \mathcal{O} = \{0\}$ is trivial. □

Theorem 1.2. *If $\mathcal{O}$ is a valuation ring of $\mathbb{K}/\mathbb{F}$ and $\mathfrak{m}$ is its maximal ideal, then it enjoys the following properties.*

(a) $\mathfrak{m}$ is a principal ideal.

(b) If t is a generator of $\mathfrak{m}$, then any $x \in \mathbb{F}$ can be represented in the form $x = ut^n$, where $u \in \mathcal{O}^\times$.

(c) In fact, $\mathcal{O}$ is a principal ideal domain; in particular, given any ideal $\mathfrak{n}$ of $\mathcal{O}$, there is an integer n such that t^n generates $\mathfrak{n}$.

Proof. We shall first need the following

Lemma 1.3. *Let $\mathcal{O}$ be a valuation ring of the number field $\mathbb{F}/\mathbb{K}$, with maximal ideal $\mathfrak{m}$. If $x \in \mathfrak{m}$ and $x_1, x_2, \ldots, x_n$ is any finite sequence of elements of $\mathfrak{m}$ such that $x_1 = x$ and $x_i \in x_{i+1}\mathfrak{m}$, then $n \leq [\mathbb{F} : \mathbb{K}(x)] < \infty$.*

PROOF. The claim is obvious in the cases of interest to Number Theory, namely in Dedekind domains, since Dedekind domain are assumed Noetherian. However, there is also a general proof. The second part of the proposition, $[\mathbb{F} : \mathbb{K}(x)] < \infty$, is trivially true since any element $z \in \mathbb{F}$ is transcendental over $\mathbb{K}$ if and only if $[\mathbb{F} : \mathbb{K}[z]] < \infty$. For the first part, we need to show that $x_1, x_2, \ldots, x_n$ are linearly independent over $\mathbb{K}(x)$. Assume the contrary; then there is a non-trivial relation

$$\sum_{i=1}^{n} a_i x_i = 0,$$

in which the coefficients a_i may be assumed polynomials (otherwise we just multiply by the least common divisor of their denominators). A further assumption we can make is that the a_i's are relatively prime; in particular, x does not divide all of them. If $b_i = a_i(0)$ are the constant terms of the coefficients, then there is a maximal $j \in \{1, 2, \ldots, n\}$ such that $b_j \neq 0$, but $b_i = 0$ for all $i > j$. We obtain

$$-a_j x_j = \sum_{i \neq j} x_i,$$

where $a_i \in \mathcal{O}$ (since $x = x_1 \in \mathfrak{m}$) and $x_i \in x_j\mathfrak{m}$ for $i < j$ by the above conditions; by the construction of j we also have $a_i = xg_i$ for $i < j$ where g_i is a polynomial in x. By the previous displayed equation we have

$$-a_j = \sum_{i<j} a_i \frac{x_i}{x_j} + \sum_{i>j} \frac{x}{x_i} g_i x_i.$$

All summands on the right hand side belong to $\mathfrak{m}$, and therefore so does a_j. On the other hand, $a_j = b_j + xg_j$ with $g_j \in \mathbb{K}[x] \subseteq \mathcal{O}$ and $x \in \mathfrak{m}$, so that $b_j = a_j - xg_j \in \mathfrak{m} \cap \mathbb{K}$, and this is absurd by theorem 1.1 (b). □

We now proceed with the proof of the claim.

(a) Assume that $\mathfrak{m}$ is not principal, and let $x_1 = x$ be any element of $\mathfrak{m}$. Since $\mathfrak{m} \neq x_1\mathcal{O}$, there is $x_2 \in \mathfrak{m} - x_1\mathcal{O}$. Since $x_2 x_1^{-1} \notin \mathcal{O}$, we must have $x_1 x_2^{-1} \in \mathcal{O}$ or $x_1 \in x_2\mathcal{O}$. Repeating the above argument, we obtain an infinite sequence that satisfies the conditions of the above lemma, a contradiction.

(b) Let $z \in \mathbb{F}$; we may even assume without loss of generality that $z \in \mathcal{O}$, since otherwise $z^{-1} \in \mathcal{O}$. If z is a unit, then $z = zt^0$, and this representation is unique. Otherwise, there is a unique maximal $m \in \mathbb{N}$, such that $z \in t^m\mathcal{O}$, hence $z = ut^m$ for some $u \in \mathcal{O}$; this follows by the lemma we showed above. Note that u must belong to $\mathcal{O}^\times$, by the construction of m.

(c) For this third claim, note that there is $n \in \mathbb{N}$, such that $n = \min\{m \mid x = ut^m, x \in \mathfrak{n}\}$. We claim that $\mathfrak{n} = t^n\mathcal{O}$, and the proof of this assertion is trivial.

□

The second class of objects that will be used in the proof of the Riemann-Roch theorem is *places*. These are defined in the following

DEFINITION 1.3. *A place P of the function field $\mathbb{F}/\mathbb{K}$ is the maximal ideal of some valuation ring $\mathcal{O}$ of $\mathbb{F}/\mathbb{K}$. Any generator t of P is called a parameter for P.*

The set of all places of $\mathbb{F}/\mathbb{K}$ will be denoted by $\mathbf{P}_{\mathbb{F}}$.

As we shall soon see, places induce and are induced by a particular class of mappings, valuations:

DEFINITION 1.4. *A valuation of a function field $\mathbb{F}/\mathbb{K}$ is a surjective homomorphism $v : \mathbb{F}^* \twoheadrightarrow \mathbb{Z}$, where $\mathbb{F}^*$ denotes the multiplicative group of nonzero elements of the field $\mathbb{F}$, such that:*

(a) $v\,|_{\mathbb{K}^*} \equiv \mathbf{0}$.

(b) (TRIANGLE INEQUALITY) $v(x + y) \geq \min\{v(x), v(y)\}$, *for any pair of* $x, y \in \mathbb{F}$.

In this context, it is convenient to formally define $v(0) = \infty$; 0 is the only element of $\mathbb{F}$ mapped to ∞.

In the above definition, the Triangle Inequality can be substituted by the following stronger version:

THEOREM 1.4. *Let v be a discrete valuation on $\mathbb{F}/\mathbb{K}$ and $x, y \in \mathbb{F}$ with $v(x) \neq v(y)$. Then, $v(x+y) = \min\{v(x), v(y)\}$.*

PROOF. Note that multiplication of $y \in \mathbb{F}$ by a scalar $a \in \mathbb{K}$ does not affect the image of y under v, namely $v(ay) = v(y)$; in particular, $v(-y) = v(y)$. Now assume without loss of generality that $v(x) < v(y)$ and suppose that $v(x+y) \neq \min\{v(x), v(y)\} = v(x)$. Then, $v(x) = v((x+y) - y) > \min\{v(x+y), v(y)\} > v(x)$, a contradiction. $\square$

To any place $P \in \mathbf{P}_{\mathbb{F}}$, we associate a mapping $v_P : \mathbb{F} \longrightarrow \mathbb{Z} \cup \{\infty\}$ by picking a prime element t of P and then letting the value of v at any $x \in \mathbb{F}$ be $m \in \mathbb{Z} \cup \{\infty\}$, the unique integer such that $x = ut^m$, for some unit u in $\mathbb{F}$. Since the representation of x in this form is unique given the prime element t the map would be well defined, if it were independent of the choice of generator t. But this is also true, for given some other prime element t', there is a unit w such that $t' = wt$, and it then becomes obvious that $x = ut^m = u't'^m$. The only problem arises when $x = 0$, but in this case we let $v_P(x) = \infty$, of course. We now have to show that this map is in fact a valuation on $\mathbb{F}/\mathbb{K}$; the only property of valuations that is not immediately obvious is the Triangle Inequality. Let $x = ut^n, y = u't^m \in \mathbb{F}$ and assume that $n \leq m < \infty$. Then, $x + y = t^n(u' + t^{m-n}u'') = t^n z$, where $z \in \mathcal{O}_P$. If $z = 0$, then $v_P(x+y) = \infty$, so there is nothing to prove; otherwise, $z = t^k u'''$, with $k \geq 0$ and $v_P(x+y) = n + k \geq n = \min\{v(x), v(y)\}$ and so the property holds.

Conversely, to any valuation v we may associate a place $P \in \mathbf{P}_{\mathbb{F}}$ defined as $P = \{z \in \mathbb{F} \mid v(z) > 0\}$; the corresponding valuation ring is $\mathcal{O}_P = \{z \in \mathbb{F} \mid v(z) \geq 0\}$. Also, we have $\mathcal{O}_P^\times = \ker v_P$, and $P = \{z \in \mathbb{F} \mid v_P(z) > 0\}$. We have the exact sequence

$$0 \longrightarrow \mathcal{O}_P^\times \xrightarrow{\iota} \mathbb{F}^* \xrightarrow{v} \mathbb{Z} \longrightarrow 0,$$

where ι and v denote the obvious inclusion and valuation.

The above discussion shows that the notions of *valuations*, *valuation rings* and *places* of a fixed function field $\mathbb{F}/\mathbb{K}$ are equivalent in the following sense: let $\mathcal{A}$ be the discrete category of all valuations, $\mathcal{B}$ the discrete category of all valuation rings and $\mathcal{C}$ the discrete category of all places of $\mathbb{F}/\mathbb{K}$. Then, the functor $\mathcal{A} \longrightarrow \mathcal{B}$ that assigns a valuation to its valuation ring, and the functor $\mathcal{A} \longrightarrow \mathcal{C}$ that assigns a valuation to its corresponding place are both invertible. Thus $\mathcal{A} \simeq \mathcal{B}$ and $\mathcal{A} \simeq \mathcal{C}$, as categories.

Let P be a place of $\mathbb{F}/\mathbb{K}$, and $\mathcal{O}_P$ its valuation ring. Since P is a maximal ideal, the residue class ring $\mathcal{O}_P/P$ is a field and we may define $\pi : \mathcal{O}_P \longrightarrow \mathcal{O}_P/P$ to be the natural projection; the domain of this homomorphism of fields can be extended to $\mathbb{F}$ by letting the image of $\mathbb{F} - \mathcal{O}_P$ be $\{\infty\}$. By theorem 1.1, we know that $\mathbb{K} \subseteq \mathcal{O}_P$ and $\mathbb{K} \cap P = \{0\}$, therefore ϕ above induces a canonical embedding of $\mathbb{K}$ into $\mathcal{O}_P/P$. This allows us to consider $\mathbb{K}$ as a subfield of $\mathcal{O}_P/P$ and repeating the construction with $\tilde{\mathbb{K}}$ instead of $\mathbb{K}$ yields the same result for $\tilde{\mathbb{K}}$. We will let $x(P) = \phi(x)$ to be the image of x under ϕ.

The *degree* of a place P is merely the degree of the extension $\mathbb{F}_P$ over $\mathbb{K}$.

We will now need a few easy facts and some definitions that will establish a dictionary between algebraic function fields and the function fields of Riemann surfaces. First, we will show the following

THEOREM 1.5. *If P is a place of $\mathbb{F}/\mathbb{K}$ and $x \in P^*$, then*

$$\deg P \leq [\mathbb{F} : \mathbb{K}(x)] < \infty.$$

PROOF. The second part of the statement is immediately seen to be true, as in theorem 1.1. The first part is equivalent to showing that given any elements $z_1, z_2, \ldots, z_n \in \mathcal{O}_P$ such that their classes $z_1(P), z_2(P), \ldots, z_n(P)$ are $\mathbb{K}$-independent, then $z_1, z_2, \ldots, z_n$ are $\mathbb{K}(x)$-independent. Assume the contrary; then there is some polynomial relation

$$\sum_{i=1}^{n} a_i z_i,$$

where the z_i are polynomials in $\mathbb{K}(x)$. Without loss of generality, we may assume that not all a_i are divisible by x, and so $a_i = b_i + x g_i$, such that not all b_i are equal to 0. But then, applying the residue class homomorphism described above, we obtain

$$\sum_{i=1}^{n} b_i z_i(P) = 0,$$

and this contradicts are assumption that $z_1(P), z_2(P), \dots, z_n(P)$ are $\mathbb{K}$-independent. $\square$

Stretching the analogy with the geometric case, we now proceed to define the meaning of *zeroes* and *poles* of elements of $\mathbb{F}/\mathbb{K}$. We have

DEFINITION 1.5. *Let $z \in \mathbb{F}_P$ and $P \in \mathbf{P}_{\mathbb{F}}$. We call P a zero of z if and only if $v_P(z) = m > 0$; in this case P is a zero of order m. We call P a pole of z if and only if $v_P(z) = m < 0$; in this case P is a pole of order $-m$.*

Our next objective will be to show that any given element of $\mathbb{F}/\mathbb{K}$ has at least one pole and at least one zero. This claim is not surprising, since intuitively $\mathbb{F}/\mathbb{K}$ is the field of meromorphic functions of the Riemann surface whose points are the places $P \in \mathbf{P}_{\mathbb{F}}$. We first prove the following

THEOREM 1.6. *Let $\mathbb{F}/\mathbb{K}$ be a function field and R a subring of $\mathbb{F}$ such that $\mathbb{K} \subseteq R \subseteq \mathbb{F}$. If $\mathfrak{a}$ is a proper ideal of R, then there exists a place $P_{\mathbb{F}}$ such that $\mathfrak{a} \subseteq P$ and $R \subseteq \mathcal{O}_P$.*

PROOF. We introduce the set $\mathcal{F} = \{S \mid S \text{ is a subring of } \mathbb{F} \text{ with } R \subseteq S \text{ and } \mathfrak{a}S \neq S\}$—by definition, $\mathfrak{a}S$ is the set of all finite sums of products $a_i s_i$, where $a_i \in \mathfrak{a}, s_i \in S$ and it is an ideal of S. We can also order $\mathcal{F}$ by inclusion. By the conditions of the theorem, $\mathcal{F}$ is not empty ($R \in \mathcal{F}$) and a typical Zorn's Lemma argument shows that $\mathcal{F}$ has a maximal element $\mathcal{O}$, that satisfies $R \subseteq \mathcal{O} \subseteq \mathbb{F}$ and $\mathfrak{a}S \neq \mathcal{O}$. It's a standard fact that $\mathcal{O}$ will be a valuation ring of $\mathbb{F}/\mathbb{K}$ (c.f. Atiyah–MacDonald, chapter 5). We repeat the proof below.

As $\mathfrak{a} \neq 0$ and $\mathfrak{a}\mathcal{O} \neq \mathcal{O}$, we have $\mathcal{O} \subsetneq \mathbb{F}$ and of course $\mathfrak{a} \subseteq \mathcal{O} - \mathcal{O}^{\times}$. Assume that there existed an element $z \in \mathbb{F}$ such that $z \notin \mathcal{O}$ and $z^{-1} \notin \mathcal{O}$; then $\mathfrak{a}\mathcal{O}[z] = \mathcal{O}[z]$ and $\mathfrak{a}\mathcal{O}[z] = \mathcal{O}[z^{-1}]$ and we may pick $a_0, a_1, \dots, a_n, b_0, b_1, \dots, b_m \in \mathfrak{a}\mathcal{O}$ such that

$$1 = a_n z^n + \cdots + a_1 z + a_0$$

and

$$1 = b_m z^{-m} + \cdots + b_1 z^{-1} + b_0.$$

We may assume without loss of generality that $m \leq n$ and then we choose m, n to be minimal; even under this assumption, clearly $m, n \geq 1$. Multiplying the two relations by $1 - b_0$ and $a_n z^n$ respectively, we obtain

$$1 - b_0 = (1 - b_0) a_n z^n + \cdots + (1 - b_0) a_1 z + (1 - b_0) a_0$$

and

$$0 = b_m a_n z^{n-m} + \cdots + b_1 a_n z^{n-1} + (b_0 - 1) a_n z^n.$$

Adding the above equations yields

$$1 = c_{n-1} z^{n-1} + \cdots + c_1 z + c_0$$

with coefficients $c_i \in \mathfrak{a}\mathcal{O}$; this is absurd by the minimality of n. Therefore, $\mathcal{O}$ is a valuation ring of $\mathbb{F}/\mathbb{K}$, as desired. $\square$

From the above, we may draw the following corollaries

COROLLARY 1.7. *Let $\mathbb{F}/\mathbb{K}$ be a function field, $z \in \mathbb{F}$ transcendental over $\mathbb{K}$. Then z has at least one zero and one pole; in particular, $\mathbf{P}_{\mathbb{F}} \neq \varnothing$.*

PROOF. Consider the ring $R = \mathbb{K}[z]$ and the ideal $\mathfrak{a} = z\mathbb{K}[z]$. The previous theorem asserts that there is a place $P \in \mathbf{P}_{\mathbb{F}}$ with $z \in P$, so that P is a zero of z. Moreover, z^{-1} has another 0, $Q \in \mathbf{P}_{\mathbb{F}}$. This Q will be a pole of z. $\square$

COROLLARY 1.8. *The field of constants $\tilde{\mathbb{K}}$ of $\mathbb{F}/\mathbb{K}$ is a finite algebraic extension of $\mathbb{K}$.*

PROOF. Since $\mathbf{P}_{\mathbb{F}} \neq \varnothing$, we may pick a place $P \in \mathbf{P}_{\mathbb{F}}$. Since $\tilde{\mathbb{K}}$ is embedded in $\mathbb{F}_P$ via the residue class map $\mathcal{O}_P \longrightarrow \mathbb{F}_P$, it follows that $[\tilde{\mathbb{K}} : K] \leq [\mathbb{F}_P : \mathbb{K}] < \infty$. $\square$

The previous corollary will prove useful in the subsequent treatment of divisors, in that it will alow us to assume that the base field is algebraically closed; this assumption can only be made when $\tilde{\mathbb{K}}$ is a finite extension of the base field $\mathbb{K}$ (since it implies that $\mathbb{F}/\tilde{\mathbb{K}}$ is also a function field).

1.2 The Weak Approximation Theorem.

In this section we introduce the Weak Approximation Theorem, which will later be used to prove that the notion of the *genus* of a function field is well-defined. The proof itself is rather technical and marginally enlightening, so it will be omitted—it is, however, readily available in [ST], Theorem I.3.1 and Iwasawa [IW], Theorem 1.1. The precise formulation of this approximation result has as follows

THEOREM 1.9 (WEAK APPROXIMATION THEOREM). *Let $\mathbb{F}/\mathbb{K}$ be a function field, $P_1, P_2, \ldots, P_n \in \mathbf{P}_{\mathbb{F}}$ pairwise distinct places of $\mathbb{F}/\mathbb{K}$, $x_1, x_2, \ldots, x_n \in \mathbb{F}$ and $r_1, r_2, \ldots, r_n \in \mathbb{Z}$. Then, there is an element $x \in \mathbb{F}$ such that*

$$v_{P_i}(x - x_i) = r_i,$$

for $1 \leq i \leq n$.

PROOF. See the above references. $\square$

A different perspective explains the name given to this proposition. If we consider the absolute value induced on $\mathbb{F}$ by the valuation v_P (we define $|z|_P = 2^{-v_P(z)}$ if $z \neq 0$ and 0 otherwise), then the Weak Approximation Theorem implies that the elements x_i can be approximated arbitrarily well by a single element $x \in \mathbb{F}$. From a different point of view, the Approximation Theorem can be regarded as the topological equivalent of the Chinese Remainder Theorem; like an approximation theorem from Number Theory, it yields information about the solution of systems of equations.

Some direct corollaries of this fact are

COROLLARY 1.10. *Any function field has infinitely many places.*

PROOF. Assume that there only finitely many places $P_1, P_2, \ldots, P_n$; by the above theorem, we can find a non-zero element $x \in \mathbb{F}$ such that $v_{P_i}(x) > 0$, for $1 \leq i \leq n$. Then x is transcendental over $\mathbb{K}$ but has no poles, a contradiction to . $\square$

THEOREM 1.11. *Let $\mathbb{F}/\mathbb{K}$ be a function field and $P_1, P_2, \ldots, P_r$ be zeroes of a fixed element $x \in \mathbb{F}$. Then*

$$\sum_{i=1}^{r} v_{P_i}(x) \deg P_i \leq [\mathbb{F} : \mathbb{K}(x_i)].$$

PROOF. See the references made in the introduction. $\square$

COROLLARY 1.12. *Any element $x \in \mathbb{F}^*$ has finitely many places and zeroes.*

PROOF. If x is constant (namely, non transcendental), then it has neither zeroes nor poles, and the claim holds trivially. Otherwise, the number of zeroes of x is bounded above by $[\mathbb{F} : \mathbb{K}(x)]$ by the previous theorem; the same argument shows that the number of zeroes of x^{-1} is finite. $\square$

1.3 Divisors.

Since the closure $\tilde{K}$ of the base field in an algebraic function field $\mathbb{F}/\mathbb{K}$ is a finite extension over $\mathbb{K}$, $\mathbb{F}/\mathbb{K}$ is also a function field, and in this case the base field is the full constant field of the function field; this situation offers certain technical advantages over the more general one, and henceforth we will assume that the base field is algebraically closed in the extension $\mathbb{F}/\mathbb{K}$.

The basic notion of this section is that of a divisor; it is given by the following

DEFINITION 1.6. *Let $Div(\mathbb{F})$ denote the free abelian group generated by the places of $\mathbb{F}/\mathbb{K}$. The elements D of $Div(\mathbb{F})$ are called divisors and can be presented as formal sums*

$$D = \sum_{P \in \mathbf{P}_{\mathbb{F}}} n_P P,$$

where the coefficients $n_p \in \mathbb{Z}$ are 0 for almost all $P \in \mathbf{P}_{\mathbb{F}}$. We also define the support $Supp(D)$ of D to be the set of all $P \in \mathbf{P}_{\mathbb{F}}$ such that $n_P \neq 0$ in the above representation of D.

We may let $v_P : \mathrm{Div}(\mathbb{F}) \longrightarrow \mathbb{Z}$ be defined by $v_P(D) = n_P$, the P'th coefficient of D. A partial ordering $\leq$ on $\mathrm{Div}(\mathbb{F})$ can be introduced by writing $D_1 \leq D_2$ if and only if $v_P(D_1) \leq v_P(D_2)$ for all $P \in \mathbf{P}$. A divisor D is called *effective* if it is non-negative, in the sense that $D \geq 0$. Since the formal sums are finite, we have a notion of *degree* for divisors: if

$$D = \sum_{P \in \mathbf{P}_{\mathbb{F}}} n_P P,$$

then we may define

$$\deg(D) = \sum_{p \in \mathbf{P}_{\mathbb{F}}} v_P(D) \deg(P).$$

The degree map is actually a homomorphism $\mathrm{Div}(\mathbb{F}) \longrightarrow \mathbb{Z}$.

Definition 1.7. *Let $x \in \mathbb{F}^*$ and denote by Z the set of zeroes of x in $\mathbf{P}_{\mathbb{F}}$. Then, we define the zero divisor of x to be*

$$(x)_0 = \sum_{P \in Z} v_P(x) P \in \mathit{Div}(\mathbb{F});$$

analogously, if $N$is the set of poles of x, we define the pole divisor of x as $(x)_\infty = \sum_{P \in N} (-v_P(x))P$. Finally, the principal divisor of x is defined to be

$$(x) = (x)_0 - (x)_\infty = \sum_{P \in \mathbf{P}_{\mathbb{F}}} v_P(x) P.$$

IN what follows, we will he assumption that K is algebraically closed will now imply that a non-zero element $x \in \mathbb{F}$ is constant if and only if $(x) = 0$; this follows from the fact that any $z \in \mathbb{F}$ which is transcendental over $\mathbb{K}$ has at least one zero and one pole and Theorem 1.1.

Definition 1.8. *The subgroup $\mathit{Princ}(\mathbb{F}) = \{(x) \mid x \in \mathbb{F}^*\}$ of $\mathit{Div}(\mathbb{F})$ is called the group of principal divisors of $\mathbb{F}/\mathbb{K}$; the operation in this group is given by $(xy) = (x) + (y)$. The class group*

$$\mathit{Class}(\mathbb{F}) = \mathit{Div}(\mathbb{F}) / \mathit{Princ}(\mathbb{F})$$

is called the divisor class group; its elements are classes of the form $[D]$, where $D \in \mathit{Div}(\mathbb{F})$ is a representative. Equality of divisor classes $[D] = [D']$ in $\mathit{Class}(\mathbb{F})$ induces an equivalence relation $D \sim D'$ in $\mathit{Div}(\mathbb{F})$. Therefore, $D \sim D'$ if and only if the difference $D - D'$ is some principal divisor (x).

Another object we need to introduce is the following class of K-vector spaces.

Definition 1.9. *For a divisor $D \in \mathit{Div}(\mathbb{F})$, we let*

$$L(A) = \{x \in \mathbb{F}^* \mid (x) \geq -D\} \cup \{0\}.$$

Explicitly, if $v_P(D) > 0$ when and only when $P \in I$, where I is a finite subset of $\mathbf{P}_{\mathbb{F}}$ and $v_Q(D) < 0$ when and only when $Q \in J$, where J is a finite subset of $\mathit{Div}(\mathbb{F})$, then $L(D)$ contains all $x \in \mathbb{F}^*$ such that x has zeroes of order at least $-v_Q(D)$ at all elements $Q \in J$ and poles of order at most $v_Q(D)$ at all elements $P \in I$.

It's now obvious that $L(D)$ is a K-vector space, and all equivalent divisors have isomorphic corresponding vector spaces (for this second assertion, assume that $D \sim D'$ and then let $\phi : L(D) \longrightarrow L(D')$ be given by $x \mapsto xz$, where z is the fixed element of $\mathbb{F}$ such that $D = D' + (z)$).

Our next objective will be to show that $\dim_{\mathbb{K}} L(D)$ is finite for all divisors D of $\mathbb{F}/\mathbb{K}$. This is intimately connected with the fact that the **genus** of $\mathbb{F}/\mathbb{K}$, is in fact well-defined and finite.

Definition 1.10. *The genus g of a function field $\mathbb{F}/\mathbb{K}$ is defined by*

$$g = \sup\{\deg D - \dim D + 1 \mid D \in \mathit{Div}(\mathbb{F})\}.$$

Note that this definition is legitimate (the supremum in question exists) since the difference $\deg D - \dim D$ *is bounded above, by the previous theorem.*

The genus is obviously non-negative; we see this by letting $D = 0$ be the trivial divisor. In fact, for any divisor D, $g \geq \deg D - \dim D + 1$ and this is called *Riemann's Inequality* (of course, Riemann had given an equivalent definition of genus and the above result was not a tautology in his approach). What is not a tautology even in our approach is that there is an integer c such that $\dim D = \deg D + 1 - g$ when $\dim D \geq c$. Indeed, pick a divisor $D_0 \in \mathrm{Div}(\mathbb{F})$ such that $g = \deg D_0 - \dim D_0 + 1$ and let $c = \deg D_0 + g$. If $\deg D \geq c$, then $\dim(D - D_0) \geq \deg(D - D_0) + 1 - g \geq c - \deg D_0 + 1 - g \geq 1$, and so there is a non-zero element $z \in L(D - D_0)$. Consider the divisor $D' = D + (z)$, which is $\geq D_0$. We observe that $\deg D - \dim D = \deg D' - \dim D' \geq \deg D_0 - \dim D_0 = g - 1$. Hence $\dim D \leq \deg D + 1 - g$, which provides the desired equality.

In order to accomplish our second goal, we need to go through some technical lemmas first, most of which involve the *dimension* $\dim D$ of a divisor D, namely the dimension $\dim L(D)$ of the associated vector space. We begin with the following

Lemma 1.13. *Let D and E be divisors of a function field $\mathbb{F}/\mathbb{K}$ with $D \leq E$. Then $L(D) \subseteq L(E)$ and also $\dim L(E)/L(D) \leq \deg E - \deg D$.*

Proof. The first part of the claim is trivial. For the second part, we may assume without loss of generality that $E = D + P$ for some $P \in \mathbf{P}_{\mathbb{F}}$, since the general case then follows by an inductive argument. Pick $t \in \mathbb{F}$ such that $v_P(E) = v_P(D) = v_P(D) + 1$. For $x \in L(E)$ we have $v_P(x) \geq -v_P(E) = -v_P(t)$, so $xt \in \mathcal{O}_P$. Thus we obtain a $\mathbb{K}$-linear map $\phi : L(E) \longrightarrow \mathbb{F}_P$ given by $x \mapsto (xt)P$. An element x is in the kernel of ϕ if and only if $v_P(xt) \geq 0$, or equivalently $v_P(x) \geq -v_P(D)$. Therefore, $\ker(\phi) = L(D)$ and therefore the induced mapping $\overline{\phi} : L(E)/L(D) \longrightarrow \mathbb{F}_P$ is injective. This implies, of course, that $\dim(L(E)/L(D)) \leq \dim \mathbb{F}_P = \deg E - \deg D$, as desired. $\square$

The next necessary statement is

Lemma 1.14. *For any divisor $D \in Div(\mathbb{F})$ is a finite dimensional $\mathbb{K}$-vector space. More precisely, if $D = D_+ - D_-$, with D_+ and D_- positive and negative divisors respectively, then* $\dim L(D) \leq \deg D_+ + 1$.

Proof. Since $L(D) \subseteq L(D_+)$, it is sufficient to show that $\dim L(D_+) \leq \deg D_+ + 1$. Since $D_+ \geq 0$, the previous lemma yields that $\dim(L(D_+)/L(0)) \leq \deg D_+$ and since $L(0) = \mathbb{K}$ (remember the assumption we made at the introduction of this section!), we obtain $\dim L(D_+) = \dim(L(D)/L(0)) + 1$; this completes the proof. $\square$

Our next theorem reinforces the analogy between elements of $\mathbb{F}/\mathbb{K}$ and meromorphic functions; now we have the arsenal required to show that all $x \in \mathbb{F}$ have as many poles as zeroes.

Theorem 1.15. *Any principal divisor has degree zero. More precisely, for any $x \in \mathbb{F}$,* $\deg(x)_0 = \deg(x)_\infty = [\mathbb{F} : \mathbb{K}(x)]$

Proof. Set $n = [\mathbb{F} : \mathbb{K}(x)]$ and

$$D = (x)_\infty = \sum_{i=1}^{r} -v_{P_i}(x) P_i,$$

where P_i for $1 \leq i \leq r$ are all the poles of x. Then,

$$\deg D = \sum_{i=1}^{r} v_{P_i}(x^{-1}) \deg P_i \leq [\mathbb{F} : \mathbb{K}(x)] = n$$

by the sharpened version of the Weak Approximation, Theorem 1.11. It would now clearly suffice to show that $n \leq \deg D$. Choose a basis $v_1, v_2, \ldots, v_n$ of $\mathbb{F}/\mathbb{K}(x)$ and a divisor $C \geq 0$ such that $(u_i) \geq -C$ for $1 \leq i \leq n$. We have $\dim(kD+C) \geq n(k+1)$ for all $k \geq 0$, which follows immediately from the fact that $x^i u_j \in L(kD+C)$ for $0 \leq i \leq k, 1 \leq j \leq n$. Setting $c = \deg C$ we obtain $n(k+1) \leq \dim(kD + C) \leq k \deg D + c + 1$, by our previous lemma. Therefore, $k(\deg D - n) \geq n - c - 1$ for all $k \in \mathbb{N}$; since the right-hand side of the previous equation is independent of k, the inequality can only be true if $\deg D \geq n$. This concludes the proof, since a symmetrical argument shows the same equality for $\deg(x)_0 = \deg(x^{-1})_\infty$. $\square$

We can draw some easy consequences of the above; the proofs are omitted.

Corollary 1.16. *(a) The degrees and the dimensions of any two equivalent divisors coincide.*

(b) If $\deg D < 0$*, then* $\dim D = 0$*.*

(c) For a divisor D *of degree* 0*, the following assertions are equivalent:*

(i) D *is principal*

(ii) $\dim D \geq 1$

(iii) $\dim D = 1$.

Notice now that in Lemma 1.14 we saw that $\dim D \leq 1 + \deg D$ for any effective divisor D. However, this holds for all divisors D with $\deg D \geq 0$. In order to verify this, assume that $\dim D > 0$, in which case $D \sim D'$ for some effective divisor D', the result now follows immediately, by the proved equalities $\deg D = \deg D'$ and $\dim D = \dim D'$.

The desired theorem is now attainable. We have

Theorem 1.17. *Any function field has finite genus. More precisely, there is a constant* $\gamma \in \mathbb{Z}$ *that only depends on* $\mathbb{F}/\mathbb{K}$ *and is such that*

$$\deg D - \dim D \leq \gamma,$$

for any $D \in Div(\mathbb{F})$*.*

Proof. First note that $D_1 \leq D_2$ implies that $\deg D_1 - \dim D_1 \leq \deg D_2 - \dim D_2$, by Lemma 1.13. Now fix a non-constant element $x \in \mathbb{F}$ and consider the specific divisor $B = (x)_\infty$. As mentioned before, there exists an effective divisor C_x such that $\dim(kB + C_x) \geq (k+1)\deg B$. for all non-negative integers k. On the other hand, $\dim(kB + C) \leq \dim(kB) + \dim C$, by lemma 1.13; the above inequalities combined yield

$$\dim(kB) \geq (k+1)\deg B - \deg C = \deg(kB) + ([\mathbb{F} : \mathbb{K}(x)] - \deg C),$$

hence

$$\deg(kB) - \dim(kB) \leq \gamma,$$

for all $k > 0$ and $\gamma = [\mathbb{F} : \mathbb{K}(x)] - \deg C$. We would like to show that the above displayed equation continues to hold when we substitute kB by any $A \in \mathrm{Div}(\mathbb{F})$.

For the above statement to hold, it clearly suffices to prove the following claim: *given a divisor* $A \in Div(\mathbb{F})$*, there exist divisors* A_1 *and* D *and a positive integer* k *such that* $A \leq A_1$*,* $A_1 \sim D$ *and* $D < kB$ (this statement is enough to finish off the proof, because then $\deg A - \dim A \leq \deg A_1 - \dim A_1 = \deg D - \dim D \leq \deg(kB) - \dim(kB) \leq \gamma$). To prove the claim, pick an effective divisor A_1 such that $A_1 \geq A$. Then $\dim(kB - A_1) \geq \dim(kB) - \deg(A_1) \geq \deg(kB) - \gamma - \deg A_1 > 0$, for $k \in \mathbb{N}$ sufficiently large. Therefore there is some element $z \in L(kB - A_1)$ and so we may set $D = A_1 - (z_0)$ for a fixed such z. Note that $A_1 \sim D$ and $D \leq A_1 - (A_1 - kB) = kB$, as desired. This completes the proof. □

1.4 The Proof of the Riemann-Roch Theorem.

We will now proceed to prove Roch's generalisation of Riemann's inequality, which is known as the Riemann-Roch theorem. Before stating it, we will need to introduce the technical objects that will be used in its proof. The first important concept is that of an adele.

Definition 1.11. *An adele of a function field* $\mathbb{F}/\mathbb{K}$ *is a mapping* $\alpha : \mathbf{P}_\mathbb{F} \longrightarrow \mathbb{F}$ *defined by* $P \mapsto \alpha_P$ *such that* $\alpha_P \in \mathcal{O}_P$ *for almost all places* P *of* $\mathbb{F}/\mathbb{K}$*. We may regard an adele as an element of* $\prod P$*, where the product ranges over all places* $P \in \mathbf{P}_\mathbb{F}$*.*

The associated space $A_\mathbb{F}$ that contains all the adeles of $\mathbb{F}/\mathbb{K}$ is called the *adele space* and is obviously a $\mathbb{K}$-vector space, with the algebraic structure inherited from the identification of $A_\mathbb{F}$ with $\prod_{P \in \mathbf{P}_\mathbb{F}} P$; in fact, $A_\mathbb{F}$ can be even regarded as a $\mathbb{K}$-algebra, though this observation will never be used henceforth.

Given an element $x \in \mathbb{F}$, the *principal adele of* x is the adele whose components are all equal to x; this definition agrees with the conditions imposed on adeles, since any $x \in \mathbb{F}$ has finitely many zeroes and poles, as we showed earlier. The above construction yields an embedding $\mathbb{F} \hookrightarrow A_\mathbb{F}$, and the valuations v_P of $\mathbb{F}/\mathbb{K}$

extend naturally to $A_{\mathbb{F}}$ by setting $v_P(\alpha) = v_P(\alpha_P)$, where α_P is the P'th component of α. By definition, $v_P(\alpha_P) \geq 0$ for almost all $P \in \mathbf{P}_{\mathbb{F}}$.

Adding to the sea of definitions there are already, we will define the following $\mathbb{K}$-subspaces of $A_{\mathbb{F}}$:

Definition 1.12. *For $D \in Div(\mathbb{F})$, we set $A_{\mathbb{F}}(D) = \{\alpha \in A_{\mathbb{F}} \mid v_P(\alpha) \geq -v_P(A)$ for all $P \in \mathbf{P}_{\mathbb{F}}\}$*

Our first observation about adeles, and in fact a preliminary version of the Riemann-Roch theorem, is the following:

Theorem 1.18. *For any divisor $D \in Div(\mathbb{F})$,*

$$\dim D = \deg D + 1 - g + \dim(A_{\mathbb{F}}/A_{\mathbb{F}}(D) + \mathbb{F}).$$

Proof. Note that despite the infinite dimension of $A_{\mathbb{F}}$, $A_{\mathbb{F}}(D)$ and $\mathbb{F}$, the above theorem asserts that the quotient space $A_{\mathbb{F}}/(A_{\mathbb{F}}(D) + \mathbb{F})$ is finite-dimensional over $\mathbb{K}$.

We will complete the proof after three lemmas. We begin with

Lemma 1.19. *Let $D_1, D_2 \in Div(\mathbb{F})$ be such that $D_1 \leq D_2$. Then $A_{\mathbb{F}}(D_1) \subseteq A_{\mathbb{F}}(D_2)$ and $\dim(A_{\mathbb{F}}(D_2)/A_{\mathbb{F}}(D_1)) = \deg D_2 - \deg D_1$.*

Proof. The first assertion of the lemma is obvious; it follows from the definition of the spaces involved and the condition on D_1, D_2. For the second claim, note that it is sufficient to prove it for in the case $D_2 = D_1 + P$, with $P \in \mathbf{P}_{\mathbb{F}}$, since the general case then follows by an inductive argument. Pick $t \in \mathbb{F}$ with $v_P(t) = v_P(D_1) + 1$ and consider the $\mathbb{K}$-linear map $\phi : A_{\mathbb{F}}(D_2) \longrightarrow \mathbb{F}_P$ defined by $\alpha \mapsto (t\alpha)_P(P)$. We can easily see that ϕ is in fact surjective, with kernel $A_{\mathbb{F}}(D_1)$. Therefore, $\deg D_2 - \deg D_1 = \deg P = [\mathbb{F} : \mathbb{K}(x)] = \dim(A_{\mathbb{F}}(D_2)/A_{\mathbb{F}}(D_1))$, and this implies the desired result. □

We now go on to show the following

Lemma 1.20. *As before, let $D_1, D_2 \in Div(\mathbb{F})$ be such that $D_1 \leq D_2$. Then $\dim((A_{\mathbb{F}}(D_2)+\mathbb{F})/(A_{\mathbb{F}}(D_1)+\mathbb{F})) = (\deg D_2 - \dim D_2) - (\deg D_1 - \dim D_1)$.*

Proof. We have the following exact sequence of linear transformations

$$0 \longrightarrow L(D_2)L(D_1) \xrightarrow{\sigma_1} A_{\mathbb{F}}(D_2)/A_{\mathbb{F}}(D_1) \xrightarrow{\sigma_2} (A_{\mathbb{F}}(D_2) + \mathbb{F})/((A_{\mathbb{F}}(D_1) + \mathbb{F}) \longrightarrow 0,$$

where σ_1 and σ_2 are defined in the obvious manner. We further claim that $\ker \sigma_2 \subseteq \mathrm{Im}\sigma_1$. Indeed, let $\alpha \in A_{\mathbb{F}}(D_2)$ with $\sigma_2(\alpha + A_{\mathbb{F}}(D_1)) = 0$. Then, $\alpha \in A_{\mathbb{F}}(D_1) + \mathbb{F}$, so there is some $x \in \mathbb{F}$ such that $\alpha - x \in A_{\mathbb{F}}(D_1)$. As $A_{\mathbb{F}}(D_1) \subseteq A_{\mathbb{F}}(D_2)$ we conclude that $x \in A_{\mathbb{F}}(D_2) \cap \mathbb{F} = L(D_2)$. Thus, $\alpha + A_{\mathbb{F}}(D_1) = x + A_{\mathbb{F}}(D_1) = \sigma_1(x + L(D_1))$ lies in the image of σ_1.

From the exactness of the previous sequence, we obtain that

$$\dim((A_{\mathbb{F}}(D_2)+\mathbb{F})/(A_{\mathbb{F}}(D_1)+\mathbb{F})) = \dim(A_{\mathbb{F}}(D_2)/A_{\mathbb{F}}(D_1)) - \dim(L(D_2)/L(D_1)) = (\deg D_2 - \dim D_2) - (\deg D_1 - \dim D_1),$$

as desired. □

The third and last lemma can be stated as

Lemma 1.21. *If B is a divisor such that $\dim B = \deg B + 1 - g$, then $A_{\mathbb{F}} = A_{\mathbb{F}}(B) + \mathbb{F}$*

Proof. First, observe that for $B_1 \geq B$, we have

$$\dim B_1 \leq \deg B_1 + \dim B - \deg B = \deg B_1 + 1 - g,$$

by lemma 1.13. On the other hand, $\dim B_1 \geq \deg B + 1 - g$ by Riemann's inequality; therefore, $\dim B_1 = \deg B_1 + 1 - g$. Now, let $\alpha \in A_{\mathbb{F}}$; by definition, we can find a divisor $B_1 \geq B$ such that $\alpha \in A_{\mathbb{F}}(B_1)$. By the proof of our previous lemma, $\dim((A_{\mathbb{F}}(B_1) + \mathbb{F})/A_{\mathbb{F}}(B) + \mathbb{F}) = (\deg B_1 - \dim B_1) - (\deg B - \dim B) = (g-1) - (g-1) = 0$. This yields the desired result, since it implies $A_{\mathbb{F}}(B) + \mathbb{F} = A_{\mathbb{F}}(B_1) + \mathbb{F}$ and since $\alpha \in A_{\mathbb{F}}(B_1)$, it follows that $\alpha \in A_{\mathbb{F}}(B) + \mathbb{F}$, as desired. □

Armed with the above, we can now prove our initial assertion. Consider an arbitrary divisor D. By Riemann's inequality, there exists a divisor $D_1 \geq D$ such that $\dim D_1 = \deg D_1 + 1 - g$. By lemma 3 in this proof, $A_{\mathbb{F}} = A_{\mathbb{F}}(D_1) + \mathbb{F}$ and in light of lemma 2 in this proof, we obtain

$$\dim(A_{\mathbb{F}}/A_{\mathbb{F}}(D)+\mathbb{F}) = \dim((A_{\mathbb{F}}(D_1)+\mathbb{F})/(A_{\mathbb{F}}(D)+\mathbb{F})) = (\deg D_1 - \dim D_1) - (\deg D - \dim D) = (g-1) + \dim D - \deg D,$$

which is the desired result. □

The above theorem states: for any divisor $D \in \mathrm{Div}(\mathbb{F})$, the formula

$$\dim D = \deg D + 1 - g + \dim(A_{\mathbb{F}}/A_{\mathbb{F}}(D) + \mathbb{F}) = i(D)$$

is true.

The next concept we shall use is that of *Weil differentials*. These will yield a second interpretation of the quantity $i(D)$ and are defined as

Definition 1.13. *A Weil differential of a function field $\mathbb{F}/\mathbb{K}$ is a map $\omega \in Hom_{\mathbb{K}}(A_{\mathbb{F}}, \mathbb{K})$ such that $\omega\mid_{A_{\mathbb{F}}(D)+\mathbb{F}} \equiv 0$ for some divisor $D \in Div(\mathbb{F})$. We set $\Omega_{\mathbb{F}}$ to be the set of all differentials on $\mathbb{F}/\mathbb{K}$ and moreover define $\Omega_{\mathbb{F}}(D)$ to be the collection of all differentials of $\mathbb{F}/\mathbb{K}$ that vanish on $A_{\mathbb{F}}(D) + \mathbb{F}$.*

We may regard $\Omega_{\mathbb{F}}$ as a $\mathbb{K}$-vector space (closure, though not immediately obvious, holds). As with adeles, $\Omega_{\mathbb{F}}(D)$ is a subspace of $\Omega_{\mathbb{F}}$.

The first fact we shall prove about differentials is

Lemma 1.22. *For any divisor $D \in Div(\mathbb{F})$, we have*

$$\dim \Omega_{\mathbb{F}}(D) = i(D).$$

Proof. The space $\Omega_{\mathbb{F}}(D)$ is isomorphic in a natural way to the space if linear forms on $A_{\mathbb{F}}/(A_{\mathbb{F}}(D)+\mathbb{F})$. Since $A_{\mathbb{F}}/(A_{\mathbb{F}}(D) + \mathbb{F})$ is finite dimensional of dimension $i(D)$ by theorem 1.18, we obtain the desired result. □

A simple consequence of the previous lemma is that $\Omega_{\mathbb{F}} \neq \varnothing$. Indeed, pick a divisor D of degree less than -1; then $\dim \Omega_{\mathbb{F}}(D) = i(D) = \dim D - \deg D + g - 1 \geq 1$, whence the desired result.

A morphism associated to any element $x \in \mathbb{F}$ and any differential ω on the space $A_{\mathbb{F}}$ of all adeles on $\mathbb{F}/\mathbb{K}$ is the natural one: define $(x\omega) : A_{\mathbb{F}} \longrightarrow \mathbb{K}$ by letting $(x\omega)(\alpha) = \omega(x\alpha)$. We see that $x\omega$ is again a differential of $\mathbb{F}/\mathbb{K}$; in fact, if ω vanishes on $A - \mathbb{F}(D) + \mathbb{F}$, then $x\omega$ vanishes on $A_{\mathbb{F}}(D + (x)) + \mathbb{F}$. Clearly, this construction extends the set of scalars of $\Omega_{\mathbb{F}}$ as a vector space from $\mathbb{K}$ to $\mathbb{F}$. We would now like to calculate the dimension of this new vector space. This answer is rather dull (if convenient) and is given by the next theorem.

Theorem 1.23. $\dim_{\mathbb{F}} \Omega_{\mathbb{F}} = 1$.

Proof. Choose a non-zero differential $\omega \in \Omega_{\mathbb{F}}$; this is possible by our remarks above. We have to show that this generates the whole space, or, equivalently, given $\omega_2 \in \Omega_{\mathbb{F}}$, there is $z \in \mathbb{F}$ such that $\omega_2 = \omega_1$; we can assume that $\omega_2 \neq 0$. Indeed, pick $D_1, D_2 \in \mathrm{Div}(\mathbb{F})$ such that $\omega_1 \in \Omega_{\mathbb{F}}(D_1)$ and $\omega_2 \in \Omega_{\mathbb{F}}(D_2)$. For a fixed divisor B, consider the $\mathbb{K}$-linear mappings $\phi_i : L(D_1 + B) \longrightarrow \Omega_{\mathbb{F}}(-B)$ (for $i = 1, 2$) defined by $x \mapsto x\omega_i$.

We claim that for an appropriate choice of the divisor B, we have

$$\phi_1(L(D_1 + B)) \cap \phi_2(L(D_2 + B)) \neq \{0\}.$$

Indeed, it is well-known from linear algebra that if U_1, U_2 are subspaces of a finite-dimensional vector space V, then

$$\dim(U_1 \cap U_2) \geq \dim U_1 + \dim U_2 - \dim V.$$

Now, let $B > 0$ be a divisor of degree sufficiently large, that

$$\dim(D_i + B) \geq \deg(D_i + B) + 1 - g$$

for $i = 1, 2$ – this is possible by Riemann's inequality – and set $U_i = \phi_i(L(D_i + B)) \subset \Omega_{\mathbb{F}}(-B)$. Since $\dim \Omega_{\mathbb{F}}(-B) = i(-B) = \deg B + 1 - g$, we obtain $\dim U_1 + \dim U_2 - \dim \Omega_{\mathbb{F}}(-B) = \deg(D_1 + B) + 1 - g +$

$\deg(D_2+B)+1-g-(\deg B+g-1) = \deg B+(\deg D_1+\deg D_2+3(1-g))$. The term in brackets is independent of B and therefore $\dim U_1+\dim U_2-\dim \Omega_{\mathbb{F}}(-B)>0$ if the degree of B is sufficiently large. By our little fact from linear algebra, it follows that $U_1\cap U_2\neq\varnothing$, as desired.

After we accomplished the above, the proof of our initial claim is easy to come: pick $x_1\in L(D_1+B), x_2\in L(D_2+B)$ such that $x_1\omega_1=x_2\omega_2\neq 0$; then $\omega_2=(x_1x_2^{-1})\omega_1$, as desired. □

We will now proceed to attach a natural divisor to any Weil differential of $\mathbb{F}/\mathbb{K}$. Before that, we have to prove the following

Lemma 1.24. *Let ω be a non-trivial differential of $\mathbb{F}/\mathbb{K}$. Then, there is a uniquely determined divisor $W\in M(\omega)=\{D\in Div(\mathbb{F})\mid \omega\mid_{A_{\mathbb{F}}(D)+\mathbb{F}}\equiv 0\}$ such that $D\leq W$ for any $D\in M(\omega)$.*

Proof. By Riemann's inequality, there exists a constant c dependent only on the function field $\mathbb{F}/\mathbb{K}$ such that $i(D)=0$ for all $D\in \mathrm{Div}(\mathbb{F})$ with the property $\deg D\geq c$ (see our remarks following the introduction of Riemann's inequality). Since $\dim(A_{\mathbb{F}}/(A_{\mathbb{F}}(D)+\mathbb{F}))=i(D)$, we have $\deg D<c$ for all $D\in M(\omega)$. Thus, we may pick a divisor W of maximal degree among those.

Suppose that W does not satisfy the desired property; then there exists a divisor $D_0\in M(\omega)$ such that $v_Q(D_0)>v_Q(W)$ at some place Q of $\mathbb{F}/\mathbb{K}$. We claim that $W+Q\in M(\omega)$, which will be a contradiction to the maximality of W. Indeed, consider an adele $\alpha=(\alpha)_P\in A_{\mathbb{F}}(W+Q)$. We may write $\alpha=\alpha'+\alpha''$ with

$$\alpha'_P=(1-\delta_{PQ})\alpha_P$$

and

$$\alpha''_P=\delta_{PQ}\alpha_P,$$

where the δ above is merely the Kronecker delta. Then, $\alpha'\in A_{\mathbb{F}}(W), \alpha''\in A_{\mathbb{F}}(D_0)$, hence $\omega(\alpha)=\omega(\alpha')+\omega(\alpha'')=0$. Therefore, ω vanishes on $A_{\mathbb{F}}(W+Q)+\mathbb{F}$, and our claim is true. This completes the proof of the lemma. □

In light of the above lemma, the following definition is legitimate

Definition 1.14. *The divisor (ω) of a Weil differential $\omega\neq 0$ is the unique divisor D of $\mathbb{F}/\mathbb{K}$ such that*

(a) ω vanishes on $A_{\mathbb{F}}(D)+\mathbb{F}$.

(b) If ω vanishes on $A_{\mathbb{F}}(D')+\mathbb{F}$, then $D'\leq D=(\omega)$.

Now for any place P and any $\omega,(\omega)$ as above, we define $v_P(\omega)=v_P((\omega))$. Moreover, a place P is called a zero of ω if $v_P(\omega)>0$ and a pole of ω if $v_P(\omega)>0$. A differential ω is called regular at a place P if $v_P(\omega)\geq 0$, and ω is called regular or holomorphic, if it is regular at all places $P\in\mathbf{P}_{\mathbb{F}}$.

Note now that $\Omega_{\mathbb{F}}(D)=\{\omega\in\Omega_{\mathbb{F}}\mid \omega=0 \text{ or } (\omega)\geq D\}$ and $\Omega_{\mathbb{F}}(0)=\{\omega\in\Omega_{\mathbb{F}}\mid \omega \text{ is regular }\}$. As a consequence of the fact that $\Omega_{\mathbb{F}}(D)=i(D)$ and the definition, we obtain $\dim\Omega_{\mathbb{F}}(0)=g$.

We will now need the following

Theorem 1.25.

(a) For a non-zero element $x\in\mathbb{F}$, and a non-zero differential $\omega\in\Omega_{\mathbb{F}}$, we have $(x\omega)=(x)+(\omega)$.

(b) Any two canonical divisors are equivalent.

Proof. A simple consequence of this statement is that the canonical divisors of $\mathbb{F}/\mathbb{K}$ form a class $[W]$ in the divisor class group Class($\mathbb{F}$); this divisor class is called the *canonical class* of $\mathbb{F}/\mathbb{K}$. We now proceed to prove the statement.

(a) If ω vanishes on $A_{\mathbb{F}}(D)+\mathbb{F}$, then $x\omega$ vanishes on $A_{\mathbb{F}}(D+(x))+\mathbb{F}$, as mentioned before, and therefore $(\omega)+(x)\leq(x\omega)$. Likewise, $(x\omega)+(x^{-1})\leq(\omega)$. Combining these inequalities, we obtain $(\omega)+(x)\leq(x\omega)\leq-(x^{-1})+(\omega)=(\omega)+(x)$, which obviously yields the desired equality.

(b) This second assertion follows from part (a) above and the fact that $\dim_{\mathbb{F}}\Omega_{\mathbb{F}}=1$.

□

As a final step before essentially completing the proof of Riemann-Roch, we must also show the following

THEOREM 1.26. *Let $D \in Div(\mathbb{F})$ be an arbitrary divisor and $W = (\omega)$ a canonical divisor of $\mathbb{F}/\mathbb{K}$. Then the mapping $\mu : L(W - D) \longrightarrow \Omega_{\mathbb{F}}(D)$ defined by $x \mapsto x\omega$ is an isomorphism of $\mathbb{K}$-vector spaces. In particular, $i(D) = \dim(W - D)$.*

PROOF. For $x \in L(W - D)$, we have $(x\omega) = (x) + (\omega) \geq -(W - D) + W = D$, hence $x\omega \in \Omega_{\mathbb{F}}(D)$. Therefore, μ maps $L(W - D)$ into $\Omega_{\mathbb{F}}(D)$. Clearly, μ is linear and injective; in order to show that μ is also surjective consider a Weil differential $\omega_1 \in \omega_{\mathbb{F}}(D)$. By the fact that $\Omega_{\mathbb{F}}$ has dimension 1 over $\mathbb{F}$, there is $x \in \mathbb{F}$ such that $\omega_1 = x\omega$, and since $(x) + W = (x\omega) = (\omega_1) \geq D$, we obtain $(x) \geq -(W - D)$, hence $x \in L(W - D)$ and $\omega_1 = \mu(x)$, as desired. Now, we have shown that $\dim \Omega_{\mathbb{F}}(D) = i(D)$; this implies that $i(D) = \dim(W - D)$. □

Now we are ready to reach our goal for this section. Without any effort at all, we can reach for the golden apple, namely

THEOREM 1.27 (RIEMANN-ROCH). *Let W be a canonical divisor of the function field $\mathbb{F}/\mathbb{K}$. Then, for any divisor $D \in Div(\mathbb{F})$,*

$$\dim D = \deg D + 1 - g + \dim(W - D).$$

PROOF. This is an immediate consequence of the definition of $i(D)$ and the previous theorem. □

A few obvious corollaries follow immediately:

COROLLARY 1.28. *Any canonical divisor W has degree $2g - 2$ and dimension g.*

PROOF. For $D = 0$, the Riemann-Roch theorem yields $\dim W = g$ and then $D = W$ yields $\deg W = 2g - 2$. □

COROLLARY 1.29. *If $D \in Div(\mathbb{F})$ is such that $\deg D \geq 2g - 1$, then $\dim D = \deg D + 1 - g$.*

PROOF. We have $\dim D = \deg D + 1 - g + \dim(W - D)$ for any canonical divisor W; since $\deg D \geq 2g - 1$ and $degW = 2g - 2$ (by the previous corollary), we have $\deg(W - D) < 0$. This can only occur when $\dim(W - D) = 0$, and this yields the desired result. □

2 An arithmetic theory for function fields.

Having developed some basic material about function fields, we are now going to discuss briefly and informally the utility of this algebraic point of view on function fields. The material we presented in the first section is usually introduced in a geometric manner, function fields corresponding to algebraic curves and their extensions to ramified covers of curves; however, we favoured the algebraic point of view because it leads very naturally to an arithmetic theory of function fields, which bears astonishing similarity to the classical theory number fields.

The proofs, however, are significantly easier in the function field setting than in the classical one. In fact, there are sweeping results that can be proved fairly easily in the case of function fields but are still intractable for number fields.

An example of such a theorem is the ABC conjecture of Oesterlé and Masser. In the case of rational numbers $\mathbb{Q}$, the usual form of the conjecture has as follows:

CONJECTURE 2.1 (ABC I, BY OESTERLÉ AND MASSER). *Assume that $A, B, C \in \mathbb{Z}$ are pairwise prime and satisfy the equation $A + B = C$. Given $\epsilon > 0$, there is a constant $M_\epsilon \in \mathbb{N}$ such that*

$$\max\{|A|, |B|, |C|\} \leq (\prod_{p|ABC} p)^{1+\epsilon},$$

where p in the above equation is positive and prime.

This conjecture can be adapted suitably for number fields and has far-reaching consequences, c.f. Lang [LA], Chapter IV, section 7, for a relevant discussion and a number of references.

We shall now reformulate ABC so that its function field analog becomes apparent. Before that, however, we will need a piece of ubiquitous terminology: we define the *height* of a rational number $r = m/n$ (where $(m,n) = 1$) to be $\mathrm{ht}(r) = \max\{\log(|m|), \log(|n|)\}$. Now we have

Conjecture 2.2 (ABC II). *Given $\epsilon > 0$, there is a constant m_ϵ such that for all rational numbers v and u that satisfy the equation $v + u = 1$ the following relation holds:*

$$\max ht(v), ht(u)\} \leq m_\epsilon + (1+\epsilon) \sum_{p|ABC} \log(p).$$

In the function field case, we will need a definition analogous to that of height above; let $\mathbb{F}/\mathbb{K}$ be a function field, and let $u \in \mathbb{F}^*$ be non-constant. Then, if M is the maximal separable extension of $\mathbb{K}(u)$ inside $\mathbb{F}$, we shall call the degree $[M : \mathbb{K}(u)]$ of the extension $M/\mathbb{K}(u)$ the *separable degree of* u; it will be denoted by $\mathrm{ord}_s(u)$. We now have

Theorem 2.1 (ABC). *Let $\mathbb{F}/\mathbb{K}$ be a function field. Then for $v, u \in \mathbb{F}^*$ that satisfy the equation $v + u = 1$ the following inequality holds*

$$\deg_s(v) = \deg_s(u) \leq 2g_{\mathbb{K}} + \sum_{P \in Supp(A+B+C)} \deg_{\mathbb{K}} P,$$

where A, B are the zero divisors of v and u in $\mathbb{F}$ and C is their common pole divisor.

The ABC conjecture for function fields implies the analogs of all classical number theoretic conjectures the classical ABC conjecture implies; in particular, it implies Fermat's Last Theorem for function fields.

An important result that used in the proof of ABC is the analog of the Hurwitz formula:

Theorem 2.2 (Hurwitz). *Let $\mathbb{L}/\mathbb{K}$ be a finite separable extension of function fields. Then,*

$$2g_{\mathbb{L}} - 2 = [\mathbb{L} : \mathbb{K}](2g_{\mathbb{K}} - 2) + \deg_{\mathbb{L}} D_{\mathbb{L}/\mathbb{K}};$$

more generally,

$$2g_{\mathbb{L}} - 2 \geq [\mathbb{L} : \mathbb{K}](2g_{\mathbb{K}} - 2) + \sum_{\mathfrak{p}} (e(\mathfrak{p}/P) - 1) \deg_{\mathbb{L}} \mathfrak{p},$$

where the sum is over all primes $\mathfrak{p}$ over $\mathbb{L}$ and equality occurs if and only if all ramified primes are tamely ramified.

All the undefined terms appearing above have meaning analogous to their meaning in the classical algebraic number theoretic context.

The above examples, however, appear meagre when compared to the other achievements of this arithmetic theory for function fields; among its proved theorems belong the Riemann Zeta Hypothesis (there is a natural, but advanced, proof due to Weil [WE] and an elementary one by Bombieri – this appears as an Appendix to Rosen's book [RO]), the Artin Primitive Roots Conjecture, and the Brumer-Stark Conjectures (there are two proofs, one by Tate and Deligne [TA] and another by Hayes [HA]). As every other legitimate theory, it also has open problems: the Langlands program has not been fully developed in the language of function fields (despite important work in this direction by Drinfeld in the 1970's) and the Birch and Swinnerton-Dyer conjecture remains a mystery. The methods one uses to formulate such conjectures and attain their proofs are not much different than the ones used here, and it's probable that results unproved in context other than that of a function field might be solvable in full generality by the same methods used in the function field case *mutatis mutandis.*

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